

NVIDIA Jetson Orin Nano/NX Modules Certificate of Volatility

Application Note

Document History

DA-11856-001_v01

Version	Date	Description of Change
01	May, 14, 2024	Initial release

Table of Contents

Jetson Orin Nano & NX Modules 4
Volatile Memory 4
Non-Volatile Memory 5
Mass Storage 6
USB 6
RF/RFID 7
Other Transmission Capabilities 7
Other Capabilities 7

Jetson Orin Nano & NX Modules

Table 1. Module Description

Model	Part Number
NVIDIA Jetson Orin NX 16GB	900-13767-0000-000
NVIDIA Jetson Orin NX 8GB	900-13767-0010-000
NVIDIA Jetson Orin Nano 8GB	900-13767-0030-000
NVIDIA Jetson Orin Nano 4GB	900-13767-0040-000

Volatile Memory

Does the device contain volatile memory (memory whose contents are lost when power is removed)?

YES / NO

If yes, describe the type, size, function, and steps to clear the memory below.

Table 2. Volatile Memory Description

Type	Size	User Modifiable	Function	Steps to Clear Memory
LPDDR5	16 GB	YES / NO	System memory	Power off
LPDDR5	8 GB	YES / NO	System memory	Power off
LPDDR5	4 GB	YES / NO	System memory	Power off

Non-Volatile Memory

Does the device contain non-volatile memory (memory whose contents are retained when power is removed)?

YES / NO

If yes, describe the type, size, function, and steps to clear the memory below.

Table 3. Non-Volatile Memory Description

Type	Size	User Modifiable	Function	Steps to Clear Memory
QSPI	512Mb	YES / NO	Storing bootloader partition.	1. **Put module in RCM (Force Recovery Mode); run the command on host PC:** `$ sudo ./tools/kernel_flash/l4t_initrd_flash.sh --initrd jetson-orin-nano-devkit mmcblk0p1` 2. **Login to module through ssh; on host PC:** `$ ssh root@fe80::1%usb0` 3. **Erase QSPI; run command on module:** `-bash-5.0# flash_erase /dev/mtd0 0 0`
EEPROM	256 Bytes	YES / NO	Storing module information.	1. **Boot module to Linux userspace** 2. **Erase either byte; run command on module:** `$ sudo i2cset -f -y 0 0x50 <byte_offset> <new_value>` 3. **Run command to dump data:** `$ sudo i2cdump -f -y 0 0x50`

⚠ **Warning:**

Clearing QSPI – After clearing QSPI, the device cannot be rebooted. User must re-flash the device.

Clearing EEPROM – After clearing EEPROM, the device cannot be re-flashed. To avoid data loss, back up any data before clearing it (e.g., BOARDID and BOARDSKU values). Refer to Flashing to Jetson Without Reading EEPROM for instructions on how to skip the EEPROM read during flashing and correct EEPROM content.

Mass Storage

Does the device contain mass storage memory?

YES / (NO)

If yes, describe the type, size, function, and steps to clear the memory below.

Table 4. Mass Storage Memory Description

Type	Size	User Modifiable	Function	Steps to Clear Memory
		YES / NO		

USB

Does the device accept USB input and if so, for what purpose (e.g., device firmware updates, etc.)?

(YES) / NO

If yes, describe below.

There is no USB connector on the module but the USB signals are ready and exposed at the SODIMM connector. This enables USB interfaces to USB devices or a USB connector on their carrier board. It is mainly used for device FW update and board debugging. Could also be used for USB memory stick.

RF/RFID

Does the device use RF or RFID to receive or transmit any data including remote diagnostics (e.g., cellular phone, Bluetooth, etc.)?

YES / NO

If yes, describe below.

Other Transmission Capabilities

Does the device employ any other methods of non-wired access to transmit or receive any data whatsoever (e.g., anything other than standard hard-wired TCP/IP, direct USB, or parallel connections)?

YES / NO

If yes, describe below.

Other Capabilities

Does the device employ any other method of communications such as a Modem to transmit or receive any data whatsoever?

YES / NO

If yes, describe below.

Notice

This document is provided for information purposes only and shall not be regarded as a warranty of a certain functionality, condition, or quality of a product. NVIDIA Corporation (“NVIDIA”) makes no representations or warranties, expressed or implied, as to the accuracy or completeness of the information contained in this document and assumes no responsibility for any errors contained herein. NVIDIA shall have no liability for the consequences or use of such information or for any infringement of patents or other rights of third parties that may result from its use. This document is not a commitment to develop, release, or deliver any Material (defined below), code, or functionality.

NVIDIA reserves the right to make corrections, modifications, enhancements, improvements, and any other changes to this document, at any time without notice.

Customer should obtain the latest relevant information before placing orders and should verify that such information is current and complete.

NVIDIA products are sold subject to the NVIDIA standard terms and conditions of sale supplied at the time of order acknowledgement, unless otherwise agreed in an individual sales agreement signed by authorized representatives of NVIDIA and customer (“Terms of Sale”). NVIDIA hereby expressly objects to applying any customer general terms and conditions with regards to the purchase of the NVIDIA product referenced in this document. No contractual obligations are formed either directly or indirectly by this document.

NVIDIA products are not designed, authorized, or warranted to be suitable for use in medical, military, aircraft, space, or life support equipment, nor in applications where failure or malfunction of the NVIDIA product can reasonably be expected to result in personal injury, death, or property or environmental damage. NVIDIA accepts no liability for inclusion and/or use of NVIDIA products in such equipment or applications and therefore such inclusion and/or use is at customer’s own risk.

NVIDIA makes no representation or warranty that products based on this document will be suitable for any specified use. Testing of all parameters of each product is not necessarily performed by NVIDIA. It is customer’s sole responsibility to evaluate and determine the applicability of any information contained in this document, ensure the product is suitable and fit for the application planned by customer, and perform the necessary testing for the application in order to avoid a default of the application or the product. Weaknesses in customer’s product designs may affect the quality and reliability of the NVIDIA product and may result in additional or different conditions and/or requirements beyond those contained in this document. NVIDIA accepts no liability related to any default, damage, costs, or problem which may be based on or attributable to: (i) the use of the NVIDIA product in any manner that is contrary to this document or (ii) customer product designs.

No license, either expressed or implied, is granted under any NVIDIA patent right, copyright, or other NVIDIA intellectual property right under this document. Information published by NVIDIA regarding third-party products or services does not constitute a license from NVIDIA to use such products or services or a warranty or endorsement thereof. Use of such information may require a license from a third party under the patents or other intellectual property rights of the third party, or a license from NVIDIA under the patents or other intellectual property rights of NVIDIA.

Reproduction of information in this document is permissible only if approved in advance by NVIDIA in writing, reproduced without alteration and in full compliance with all applicable export laws and regulations, and accompanied by all associated conditions, limitations, and notices.

THIS DOCUMENT AND ALL NVIDIA DESIGN SPECIFICATIONS, REFERENCE BOARDS, FILES, DRAWINGS, DIAGNOSTICS, LISTS, AND OTHER DOCUMENTS (TOGETHER AND SEPARATELY, “MATERIALS”) ARE BEING PROVIDED “AS IS.” NVIDIA MAKES NO WARRANTIES, EXPRESSED, IMPLIED, STATUTORY, OR OTHERWISE WITH RESPECT TO THE MATERIALS, AND EXPRESSLY DISCLAIMS ALL IMPLIED WARRANTIES OF NONINFRINGEMENT, MERCHANTABILITY, AND FITNESS FOR A PARTICULAR PURPOSE. TO THE EXTENT NOT PROHIBITED BY LAW, IN NO EVENT WILL NVIDIA BE LIABLE FOR ANY DAMAGES, INCLUDING WITHOUT LIMITATION ANY DIRECT, INDIRECT, SPECIAL, INCIDENTAL, PUNITIVE, OR CONSEQUENTIAL DAMAGES, HOWEVER CAUSED AND REGARDLESS OF THE THEORY OF LIABILITY, ARISING OUT OF ANY USE OF THIS DOCUMENT, EVEN IF NVIDIA HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. Notwithstanding any damages that customer might incur for any reason whatsoever, NVIDIA’s aggregate and cumulative liability towards customer for the products described herein shall be limited in accordance with the Terms of Sale for the product.

Trademarks

NVIDIA, the NVIDIA logo, are trademarks and/or registered trademarks of NVIDIA Corporation in the U.S. and other countries. Other company and product names may be trademarks of the respective companies with which they are associated.

VESA DisplayPort

DisplayPort and DisplayPort Compliance Logo, DisplayPort Compliance Logo for Dual-mode Sources, and DisplayPort Compliance Logo for Active Cables are trademarks owned by the Video Electronics Standards Association in the United States and other countries.

HDMI

HDMI, the HDMI logo, and High-Definition Multimedia Interface are trademarks or registered trademarks of HDMI Licensing LLC.

Arm

Arm, AMBA, and Arm Powered are registered trademarks of Arm Limited. Cortex, MPCore, and Mali are trademarks of Arm Limited. All other brands or product names are the property of their respective holders. ”Arm” is used to represent Arm Holdings plc; its operating company Arm Limited; and the regional subsidiaries Arm Inc.; Arm KK; Arm Korea Limited.; Arm Taiwan Limited; Arm France SAS; Arm Consulting (Shanghai) Co. Ltd.; Arm Germany GmbH; Arm Embedded Technologies Pvt. Ltd.; Arm Norway, AS, and Arm Sweden AB.

OpenCL

OpenCL is a trademark of Apple Inc. used under license to the Khronos Group Inc.

Copyright

© 2024 NVIDIA Corporation. All rights reserved.

NVIDIA Corporation | 2788 San Tomas Expressway, Santa Clara, CA 95051
http://www.nvidia.com